

Overview: DB Mirroring and Transaction Logs

In **Database Mirroring**, SQL Server maintains **two copies of the database**:

1. **Principal** – active database where users connect.
2. **Mirror** – synchronized copy of principal (can be synchronous or asynchronous).
3. **Transaction Log** – all changes on the principal are written to the log and sent to the mirror.

Important:

- The **log file** on the principal **cannot truncate** until the mirror confirms the transaction.
- If the mirror is unavailable, slow, or disconnected, **log growth can occur** → log file fills up → transactions may fail.

Common Reasons for Database Mirroring Log File Full

#	Cause	Description	Real-Time Scenario
1	Mirror unavailable / disconnected	Mirror server is offline, network issues, or mirroring session broken.	DB transactions on principal fail; log grows indefinitely.
2	Mirror latency	Synchronous mode: mirror cannot keep up due to heavy load or slow I/O.	Log accumulation on principal; delayed commits for synchronous mode.
3	Long-running transactions	Large batch updates, index rebuilds, or bulk operations.	Log cannot truncate until transaction completes → rapid growth.
4	Checkpoints not happening	Log truncation depends on database recovery; if DB has pending checkpoint, log stays.	High I/O waits, slow log truncation.
5	Log backup not performed (for FULL recovery)	Even with mirroring, log may grow if regular backups are not done.	Scheduled jobs fail → disk space exhausted.
6	Mirroring snapshot delay	Mirror database snapshot for reporting blocks log truncation.	Reports query snapshot → prevents truncation.
7	Network bottleneck	Slow network causes log send queue to increase.	log_send_queue grows → log fills.
8	Database in high DML workload	Frequent inserts/updates/deletes.	Synchronous mirroring + high DML → log grows quickly.
9	Transaction log autogrowth settings	Autogrowth not enabled or limited max size.	Log full error, failed transaction.

Symptoms of Mirroring Log File Full

- SQL Error: The transaction log for database 'DBName' is full.
- sys.database_mirroring shows **safety_mode = FULL** (synchronous) and **mirroring_state_desc = SUSPENDED**.
- Log_send_queue > 0: check with:

```
SELECT name, log_send_queue_size, redo_queue_size, mirroring_state_desc
FROM sys.database_mirroring;
```
- Disk usage: Transaction log file (.ldf) grows to max size.
- Application errors or slow performance due to blocked transactions.

Resolution Steps in Detail

1 Check Mirroring Status

```
SELECT DB_NAME(database_id) AS DBName,
       mirroring_state_desc,
       mirroring_safety_level_desc,
       log_send_queue_size, redo_queue_size
FROM sys.database_mirroring;
```

Interpretation:

- SUSPENDED → mirror unavailable → log cannot truncate.
- High log_send_queue_size → network/mirror lag.

2 Resume Mirroring (if Suspended)

`ALTER DATABASE [DBName] SET PARTNER RESUME;`

- After resume, log sends to mirror → truncation resumes.
- **Real-time scenario:** Mirror VM was offline for patching → resume restores mirroring.

3 Check Transaction Log Usage

`DBCC SQLPERF(LOGSPACE);`

- Shows log usage % per database.
- If > 80–90% → risk of log full.

4 Free up Log Space

a) Backup Transaction Log (FULL Recovery Model)

`BACKUP LOG [DBName] TO DISK = 'D:\Backups\DBName_Log.trn';`

- Truncates inactive part of log.
- **Use case:** Daily log backups scheduled for high-transaction DB in mirroring setup.

b) Shrink Log (if necessary)

`DBCC SHRINKFILE('DBName_Log', 1024); -- target size in MB`

- Only if disk space is critical.
- Shrink is **not a routine operation**; plan for log growth.

5 Check for Long-running Transactions

```
SELECT session_id, status, start_time, command,
       total_elapsed_time/1000 AS Duration_s
FROM sys.dm_exec_requests
WHERE database_id = DB_ID('DBName')
ORDER BY total_elapsed_time DESC;
```

- **Resolve:** Kill unnecessary transactions or let long batch finish.
- **Use case:** Bulk ETL jobs blocked log truncation → monitor batch schedules.

6 Monitor Network and Mirror I/O

```
SELECT name, log_send_queue_size, redo_queue_size
FROM sys.database_mirroring;
```

- **Resolution:**
 - Check network latency between principal and mirror.
 - Ensure mirror has fast disks and sufficient CPU/memory.

7 Enable Autogrowth (if needed)

`ALTER DATABASE [DBName] MODIFY FILE (NAME = N'DBName_Log', MAXSIZE = UNLIMITED, FILEGROWTH = 512MB);`

- Prevents “log full” errors during high-transaction periods.
- **Use case:** OLTP DB with synchronous mirroring and bursts of transactions.

8 Temporary Resolution (Unsynced Mirror)

- If mirror is **offline and cannot be restored immediately**:
`ALTER DATABASE [DBName] SET PARTNER OFF;`
- **Warning:** Mirroring stops; principal continues, log truncation resumes.
- Once mirror is back → re-establish mirroring.

Real-world use: Short-term workaround for critical OLTP systems to avoid application downtime.

⚡ Real-Time Use Cases

1. High DML OLTP System:

- Daily bulk imports → log fills quickly.
- **Resolution:** Schedule log backups during off-peak hours, increase log file, monitor mirror latency.

2. Network Disruption Between Sites:

- Synchronous mirror over WAN → log_send_queue grows.
- **Resolution:** Resume mirroring after network restored or temporarily suspend to allow principal log truncation.

3. Long-Running Batch Jobs:

- Index rebuild or ETL → log fills and mirror falls behind.
- **Resolution:** Break batch into smaller chunks, monitor log_send_queue.

Best Practices to Prevent Mirroring Log Full Issues

Practice	Description
Regular Transaction Log Backups	Keeps log truncated and disk space under control.
Monitor Mirroring State	sys.database_mirroring and alerts for SUSPENDED state.
Monitor Log Growth	Set autogrowth with reasonable increments; pre-size log file.
Network Monitoring	Ensure fast and reliable connectivity between principal/mirror.
Batch Size Management	Large ETL or bulk ops should be in chunks to avoid log spikes.
Alerting	SQL Agent job for log > 80% usage.

Summary

- **Root causes:** Mirror offline, slow mirror, network issues, long-running transactions, heavy DML, missing log backups, checkpoint delays.
- **Symptoms:** Log grows rapidly, PAGEIOLATCH waits, blocked transactions, application errors.
- **Resolutions:** Resume mirroring, log backup, shrink if needed, check long-running queries, increase log file size, monitor mirror status.
- **Best practice:** Automate log backups, monitor sys.database_mirroring, pre-size log, maintain network stability.

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